Topic to Discuss

→ Double Integration Using Trapezoidal Rule

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Double Integration using Trapezoidal Rule

Suppose,

$$I = \int_c^d \int_a^b f(x,y)\,dx\,dy$$

If each interval is divided into 2 equal parts,

then

$$h = \frac{b-a}{2} \quad , \quad K = \frac{d-c}{2}$$

Then formula is :

$$\int_c^d \int_a^b f(x,y)\,dx\,dy = \frac{hK}{4}\left[f_{00} + f_{02} + f_{20} + f_{22} + 2\left(f_{01} + f_{10} + f_{12} + f_{21}\right) + 4f_{11}\right]$$

where,

$$f_{ij} = f(x_i, y_j)$$

So,

corner points have weight 1

edge middle point have weight 2

centre point have weight 4

In short,

$$I = \frac{hk}{4}\left[\text{corners} + 2(\text{edge points}) + 4(\text{interior points})\right]$$

corner	edge	corner
edge	middle/interior	edge
corner	edge	corner

For composite Trapezoidal Rule :

Now, the interval in x-direction is divided into n equal parts and in y-direction into m equal parts, then

$$h = \frac{b-a}{n} \quad , \quad K = \frac{d-c}{m}$$

The formula is :

$$I = \frac{hK}{4}\left[\sum_{\text{corners}} f(x_i, y_j) + 2\sum_{\text{edge}} f(x_i, y_j) + 4\sum_{\text{interior}} f(x_i, y_j)\right]$$

Where,

$$\sum_{\text{Corner}} f(x_i, y_j) = \sum_{i=1}^{n-1}\sum_{j=1}^{m-1} f(x_i, y_j)$$

Numerical Problem :

Q: Evaluate the following double Integration

$$\int_{-2}^{2}\int_{0}^{4}(x^2-xy+y^2)\,dx\,dy$$

using trapezoidal rule.

Solution: let $y = f(x,y) = x^2 - xy + y^2$

let h is the step length along x-direction

and K is step length along y-direction,

also considered $n=4$ & $m=4$

$\therefore$ $h = \dfrac{b-a}{n}$ & $K = \dfrac{d-c}{m}$

$= \dfrac{4-0}{4} = 1$ $\qquad = \dfrac{2-(-2)}{4}$

$= 1$

Now lets make table, $y = x^2 - xy + y^2$

y \ x	0	1	2	3	4
-2	4	7	12	19	28
-1	1	3	7	13	21
0	0	1	4	9	16
1	1	1	3	7	13
2	4	3	4	7	12

Apply trapezoidal rule formula,

$$\int_{-2}^{2}\int_{0}^{4} f(x)\,dx = \frac{hk}{4}\left[\sum_{\text{corners}} f(x_i, y_j) + 2\sum_{\text{edge}} f(x_i, y_j) + 4\sum_{\text{interior}} f(x_i, y_j)\right]$$

$$= \frac{1\times 1}{4}\Big[(4+28+4+12) + 2(7+12+19+1+0+1+3+4+7+21+16+13)$$
$$+ 4(3+7+13+1+4+9+1+3+7)\Big]$$

$$= \frac{1}{4}\left[48 + 208 + 192\right]$$
$$= 112$$

Ans.

Homework Problem :

Q: Solve the integral using trapezoidal rule.

$$\int_0^1 \int_0^1 e^{x+y} \, dx \, dy. \quad \text{Use } h = K = 0.5$$

Solution: Let, $f(x) = e^{x+y}$

we have,

$h = K = 0.5$

Table for $y = e^{x+y}$

y \ x	0	0.5	1
0	0	1.648721	2.71828182
0.5	1.648721	2.71828182	4.481689
1	2.71828	4.481689	7.389056

Now Applying trapezoidal rule,

$$\int_0^1 \int_0^1 f(x)\,dx = \frac{hk}{4}\left[\sum_{\text{corners}} f(x_i, y_j) + 2 \sum_{\text{edge}} f(x_i, y_j) + 4 \sum_{\text{interior}} f(x_i, y_j) \right]$$

$$= \frac{0.5 \times 0.5}{4} \Big[0 + 2.718282 + 2.718282 + 7.389056$$
$$+ 2(1.648721 + 1.648721 + 4.481689 + 4.481689)$$
$$+ 4(2.718282) \Big]$$

$$= 3.0762 \text{ Ans.}$$

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